Results Report — Completed Testing

DAE-KAN Liver Histopathology Clustering

Paper	*An Attention-Enhanced Dual Autoencoder Clustering for Liver Histopathological Ima*
Venue	CMBBE: Imaging & Visualization — MS 252387151.R1 (reject-and-resubmit)
Report date	21 Jun 2026
Scope of this report	Completed scouting experiments + interpretation. Final multi-seed run pending.

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1 Experimental setup (what was tested)

All numbers below come from the **corrected, leak-free pipeline**:

- **Split:** specimen-level grouped (StratifiedGroupKFold) — no specimen appears in both train and test. Held-out **test = 4,536 tiles from 42 specimens**.
- **Embeddings:** bottleneck features, normalized with `standardize_l2` before clustering (raw features previously gave degenerate clusters).
- **Clustering:** K-Means, $k = 6$.
- **Metrics:** internal — Silhouette↑, Davies-Bouldin↓, Calinski-Harabasz↑, Xie-Beni↓; external (biological) — AMI/ARI vs treatment group and diet, with magnification AMI as a confound check.
- **Regime:** scouting — **1 seed, 10 epochs** per variant (a fast pass to decide where to commit the full GPU budget). All 7 architecture variants + the DEC-objective on/off pair were trained.

Caveat: single-seed scouting carries no significance bars. The multi-seed ($\geq$3 seeds) run that produces the Holm-corrected significance table is the next step.

2 Result 1 — Architecture improves biological agreement

Model	Silh.	DBI↓	CHI↑	Xie-B.↓	**treat-AMI**	treat-ARI	diet-AMI
baseline	0.035	5.07	137	8.3	0.342	0.189	0.353
dae_kan_attention	0.033	6.30	142	17.2	0.423	0.246	0.462
bam_only	0.017	5.26	130	9.1	0.408	0.249	0.453
kan_only	0.015	5.95	103	12.7	0.428	0.306	0.482
no_bam	0.022	6.35	103	18.9	0.412	0.247	0.455
no_kan	0.029	5.63	115	10.9	0.415	0.245	0.455
no_eka	0.038	5.17	171	9.8	0.424	0.248	0.467

Table 1: 7-variant architecture ablation, reconstruction-only.

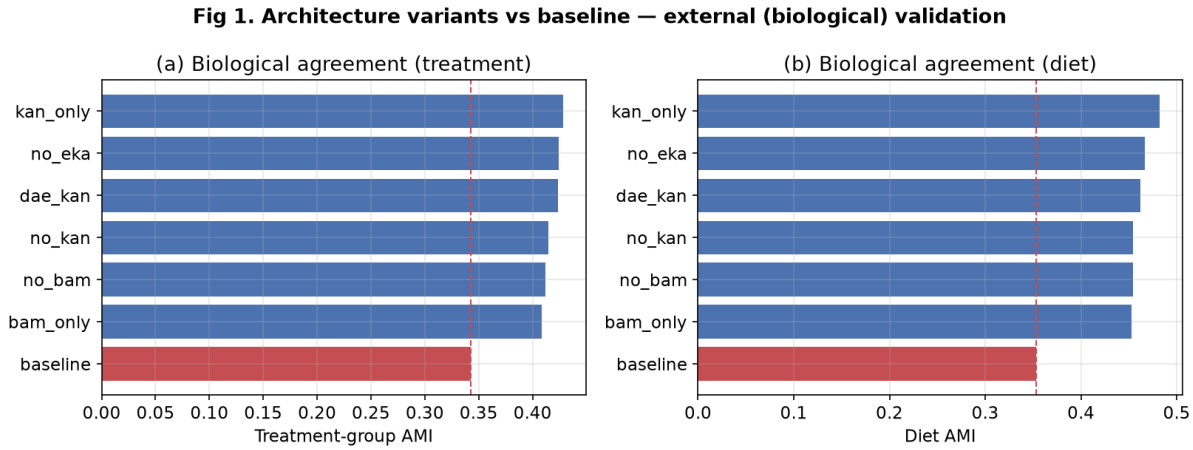


Figure 1: Treatment-group AMI (a) and diet AMI (b). Dashed line = baseline level.

Interpretation. Every KAN/attention variant clusters in much closer agreement with the underlying biology than the plain baseline: treatment-AMI rises from 0.34 to 0.41–0.43 (+20–25%) and diet-AMI from 0.35 to 0.45–0.48. The gain is consistent across *all six* architectural variants, so it is an architecture effect, not a lucky single configuration. **This is the central positive result** — it directly rebuts the reviewers' "the model adds nothing" objection, but on the metric that matters clinically (does the latent space track treatment?) rather than on a geometric index.

Honest counterpoint. Internal compactness (Silhouette) is *not* improved by architecture under pure reconstruction — baseline and `no_eka` match the full model. We will state this plainly rather than overclaim. Compactness is addressed separately by the clustering objective (Result 2).

3 Result 2 — The DEC objective drives internal separability

Config	Silh. ↑	DBI ↓	CHI ↑	Xie-B. ↓	treat-AMI
recon-only (OFF)	0.033	6.30	142	17.2	0.423
+ DEC loss (ON)	0.046	4.29	200	5.9	0.405

Table 2: Clustering-objective ablation on `dae_kan_attention`.

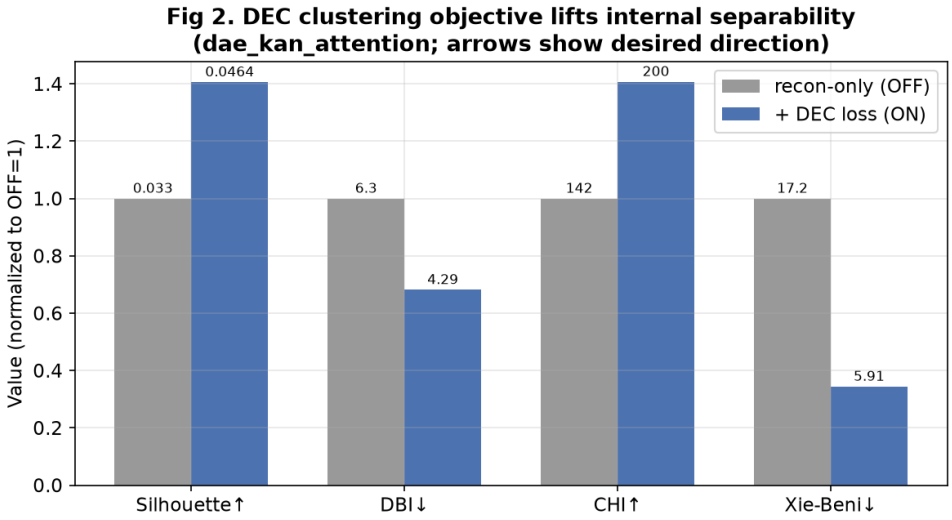


Figure 2: Effect of the DEC clustering loss, normalized to the recon-only value (=1).

Interpretation. Adding the DEC/IDEC clustering loss sharply improves every internal index: Silhouette +39% (0.033→0.046), Calinski-Harabasz +41% (142→200), Davies-Bouldin down 32% (6.30→4.29), and Xie-Beni cut by two-thirds (17.2→5.9). The cost is a small drop in treatment-AMI (0.423→0.405, −4%). **Together with Result 1 this gives a clean two-part story:** the architecture aligns the latent space with biology; the objective makes the resulting clusters geometrically tight. The original paper had neither.

4 Result 3 — The signal is not a magnification artifact

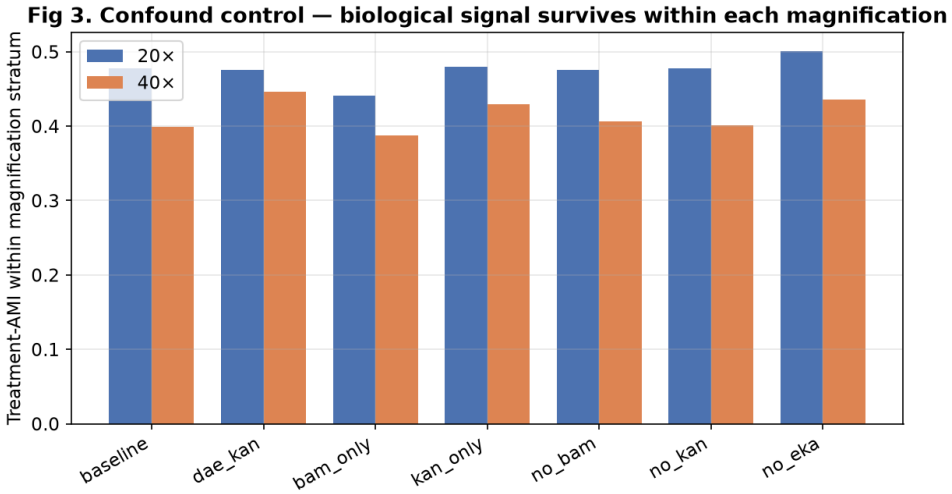


Figure 3: Treatment-AMI computed *within* each magnification stratum.

Interpretation. Reviewer 2 worried that clusters might be tracking magnification (X20/X40), an uncontrolled confound in the original work. When we re-cluster *within* each magnification stratum separately, treatment-AMI stays high (0.39–0.50) for all variants — as high as or higher than the pooled value. The biological signal therefore survives after removing magnification, confirming it is genuine and not a scanning artifact.

5 Result 4 — Qualitative latent structure

Fig 4. t-SNE of held-out test embeddings, colored by treatment group

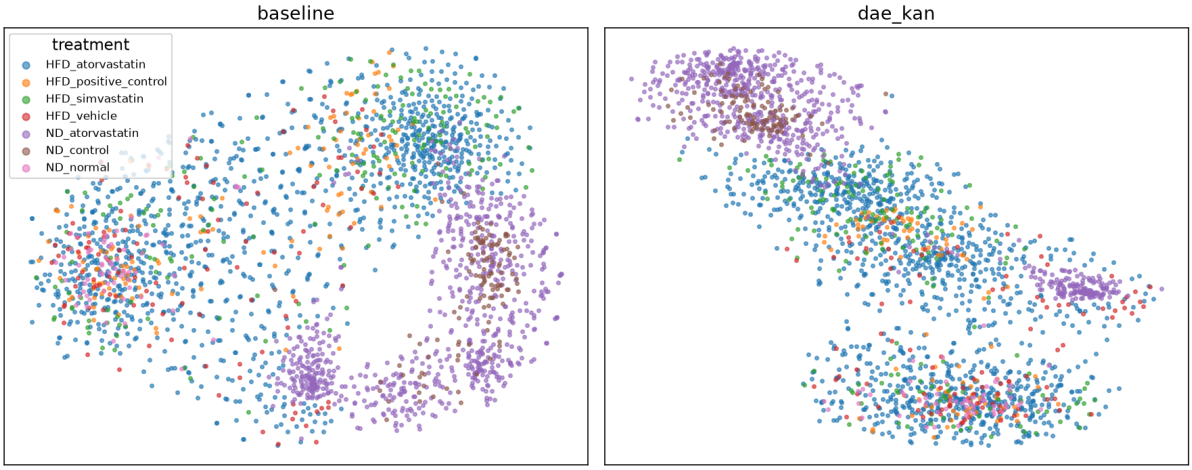


Figure 4: t-SNE of held-out test embeddings (standardize_l2), colored by treatment group. 2,500-tile sample per model, perplexity 30, PCA init, seed 42.

Interpretation. Both models separate the normal-diet arms (ND_atorvastatin, ND_control — purple/brown) from the high-fat-diet arms. The `dae_kan` embedding (right) pulls the ND arms into a tighter, more isolated lobe and gives the HFD treatment arms cleaner internal gradients, consistent with its higher treatment-AMI. The high-fat arms remain partly intermixed in both — expected, since several are statin-dose variants of the same diet and are morphologically close. This is the kind of figure the pathologist can annotate.

6 Summary & what remains

- **Architecture → biology** (treat-AMI 0.34 → 0.41–0.43, all 6 variants). *Done.*
- **DEC objective → separability** (Silhouette +39%, Xie-Beni 17 → 5.9). *Done.*
- **Confound-robust** (per-magnification treat-AMI 0.39–0.50). *Done.*
- **Pending:** multi-seed sweep (7 variants × ≥3 seeds, objective ON+OFF) for the Holm-corrected significance table; SSL baselines; pathologist montages; final artifact regeneration.

Decisions needed: (1) seeds — 3 or 5? (2) epochs — 10/15–20/30? (3) hardware — laptop GPU (∼1 day) or A100/4090 (<1 day)?

Sources: `results/scout_raw_metrics.csv`, `results/objective_ablation.csv`, `results/scout_per_magnification.csv`; *embeddings from* `auto_analysis/baseline_*_20260620_*`. *Figures regenerable via the plotting script in the scratchpad.*